

## Special Techniques in Electrostatics

**Coulomb's Law (tedious in terms of vector nature & the evaluation of integral)**

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\hat{r}}{r^2} \rho(\vec{r}') d\tau'$$

**Gauss's Law (Only when symmetry permits)**

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

**Calculation of potential (to simplify the procedure)**

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{1}{r} \rho(\vec{r}') d\tau'$$

$$\vec{E} = -\vec{\nabla} V$$

**Still, one needs to evaluate the tedious integral.**

**Further, if the conductors are involved, situation is more complicated as  $\rho$  itself may not be known in advance!**

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## Special Techniques in Electrostatics

**We need to find a simpler, general, accurate and powerful technique to calculate EF for any charge distribution**

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad \text{Poisson's Equation}$$

In the region, where there is no charge,

$$\nabla^2 V = 0 \quad \text{Laplace's Equation}$$

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0 \quad \vec{\nabla} \times \vec{E} = 0 \quad \vec{E} = -\vec{\nabla} V$$

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## Special Techniques in Electrostatics

**We need to find a simpler, general, accurate and powerful technique to calculate EF for any charge distribution**

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad \text{Poisson's Equation}$$

$$\nabla^2 V = 0 \quad \text{Laplace's Equation}$$

**PE/LE itself does not determine  $V$ .**

**Example:** 
$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

**One dimension problem:** 
$$\frac{d^2 V}{dx^2} = 0 \quad y = mx + n$$

**We need two appropriate boundary conditions to get the solution.**

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## Special Techniques in Electrostatics

**We need to find a simpler, general, accurate and powerful technique to calculate EF for any charge distribution**

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad \text{Poisson's Equation}$$

$$\nabla^2 V = 0 \quad \text{Laplace's Equation}$$

**PE/LE itself does not determine  $V$ .**

**Example:** 
$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

**Two-dimension problem:** 
$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

**Things are very complicated here.**

**Problems in 3D are even more complicated!**

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## Special Techniques in Electrostatics

We need to find a simpler, general, accurate and powerful technique to calculate EF for any charge distribution

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad \text{Poisson's Equation}$$

$$\nabla^2 V = 0 \quad \text{Laplace's Equation}$$

PE/LE itself does not determine  $V$ . What we need in addition is an appropriate boundary conditions.

What are those appropriate BCs, sufficient to determine  $V$ ?

Does PE/LE along with appropriate BC lead to the unique solution for the EF even in the presence of conductor?

YES

Uniqueness Theorem

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## Special Techniques in Electrostatics

### Uniqueness Theorem

$$\nabla \cdot \mathbf{E}_1 = \frac{1}{\epsilon_0} \rho \quad \nabla \cdot \mathbf{E}_2 = \frac{1}{\epsilon_0} \rho$$

$$\mathbf{E}_3 \equiv \mathbf{E}_1 - \mathbf{E}_2,$$

$$\nabla \cdot \mathbf{E}_3 = 0$$

$$\nabla \cdot (V_3 \mathbf{E}_3) = V_3 (\nabla \cdot \mathbf{E}_3) + \mathbf{E}_3 \cdot (\nabla V_3) = -(E_3)^2.$$

$$\int_V \nabla \cdot (V_3 \mathbf{E}_3) d\tau = \oint_S V_3 \mathbf{E}_3 \cdot d\mathbf{a} = - \int_V (E_3)^2 d\tau$$

$$\int_V |\mathbf{E}|^2 d\tau = - \oint_S V \vec{\mathbf{E}} \cdot d\vec{\mathbf{a}}$$

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## Special Techniques in Electrostatics

Answer: Uniqueness Theorem

$$\int_V |\mathbf{E}|^2 d\tau = - \oint_S V \vec{\mathbf{E}} \cdot d\vec{\mathbf{a}} \quad \vec{\mathbf{E}} = \vec{\mathbf{E}}_1 - \vec{\mathbf{E}}_2 \quad V = V_1 - V_2$$

Any BC, which makes the surface integral zero, makes  $\vec{\mathbf{E}} = 0$  and hence the solution unique.

**Note:** We do not know the region ( $S$  and  $V$ ) of integration

We do not know, what are specific BCs

What we know is, BC should be such that the surface Integral must vanish

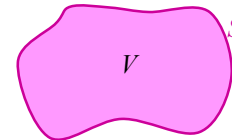
This only is sufficient to prove that the solution of PE/LE are unique!

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## Special Techniques in Electrostatics

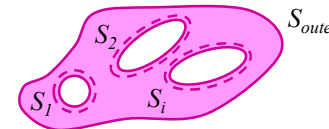
Region ( $S$  and  $V$ ) of Integration:

A: In the absence of conductor but no cavity



$S$  can be taken to infinity as well. Then integration is over all space and  $V$  generally taken to be zero there.

B: In the absence of conductor but having cavity



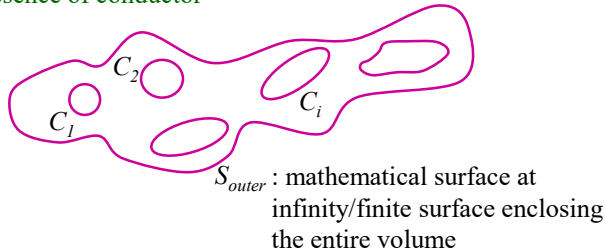
$$\int_V |\mathbf{E}|^2 d\tau = - \sum_i \oint_{S_i} V \vec{\mathbf{E}} \cdot d\vec{\mathbf{a}}$$

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## Special Techniques in Electrostatics

Region ( $S$  and  $V$ ) of Integration:

A: In the presence of conductor



$$\int_V |\vec{E}|^2 d\tau = -\sum_i \oint_{S_i} V \vec{E} \cdot d\vec{a}$$

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## Special Techniques in Electrostatics

**Boundary Conditions:**

$$\int_V |\vec{E}|^2 d\tau = -\sum_i \oint_{S_i} V \vec{E} \cdot d\vec{a}$$

BC to make surface integration vanish is that  $V$  vanishes on all surfaces  $S_i$

→  $V_1 = V_2 = \dots$  on each surface

→ **Potential is specified on the surface**

**Dirichlet Boundary Conditions**

BC to make surface integration vanish is that the normal component of  $\vec{E}$  ( $= \hat{n} \cdot \vec{E}$ ) vanishes on all surfaces  $S_i$

→  $\vec{E}^\perp_1 = \vec{E}^\perp_2 = \dots$  on each surface

→ **Normal component of the EF is specified on the surface**

**Neumann Boundary Conditions**

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## Special Techniques in Electrostatics

**First Uniqueness Theorem of ED:**

$$\int_V |\vec{E}|^2 d\tau = -\sum_i \oint_{S_i} V \vec{E} \cdot d\vec{a}$$

The EF in a volume  $V$  is uniquely defined by the solution of PE, if either the potential or its normal derivative (directional derivative,  $\vec{E}^\perp = \vec{E} \cdot \hat{n} = -\hat{n} \cdot \vec{\nabla} V$ ) is specified on each surface of the volume.

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## Special Techniques in Electrostatics

**Second Uniqueness Theorem of ED (in the presence of conductor):**

So far surface of volume  $V$  could be arbitrary surface on which BC was specified.

If any surface is the surface of a conductor, that must be an equipotential with constant value of  $V$  over the surface of the conductor.

$$\int_V |\vec{E}|^2 d\tau = -\sum_i \oint_{S_i} V_i \vec{E} \cdot d\vec{a}$$

Means, the normal component of  $\vec{E}$  must vanish →  $\vec{E}^\perp_1 = \vec{E}^\perp_2 = \dots$  on each surface → Normal component of the EF is specified on the surface.

But,  $E^\perp = \sigma / \epsilon_0$ . Thus, evaluation of  $\oint_S \vec{E} \cdot d\vec{a}$  is equal to the net charge on the conductor.

EF in a volume  $V$  is uniquely defined by the solution of PE, if either the potential or the net charge is specified on each conductor.

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## Special Techniques in Electrostatics

### Second Uniqueness Theorem of ED (in the presence of conductor):

If any surface is the surface of a conductor, that must be an equipotential with constant value of  $V$  over the surface of the conductor.

$$\int_V |\vec{E}|^2 d\tau = -\sum_i V_i \oint_{S_i} \vec{E} \cdot d\vec{a}$$

Means, the normal component of  $\vec{E}$  must vanish  $\rightarrow \vec{E}^\perp_1 = \vec{E}^\perp_2 = \dots$  on each surface  $\rightarrow$  Normal component of the EF is specified on the surface.

But,  $E^\perp = \sigma / \epsilon_0$ . Thus, evaluation of  $\oint_S \vec{E} \cdot d\vec{a}$  is equal to the net charge on the conductor.

EF in a volume  $V$  is uniquely defined by the solution of PE, if either the potential or the net charge is specified on each conductor.

**Note:** Actual surface charge distribution can not be known for a conductor until the problem is solved. Hence Neumann BC will not be a useful BC for conductors.

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## Special Techniques in Electrostatics

### Consequence of Uniqueness Theorem of ED

Any method of solution is valid as long as the resulting potential satisfies PE/LE and the boundary conditions.

Even guessing a solution would be alright if it works.

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## Special Techniques in Electrostatics

### Application (How to solve a problem)

#### What we need...

- 🍁 Appropriate DE (PE/LE)
- 🍁 Region
- 🍁 Appropriate Boundary Conditions

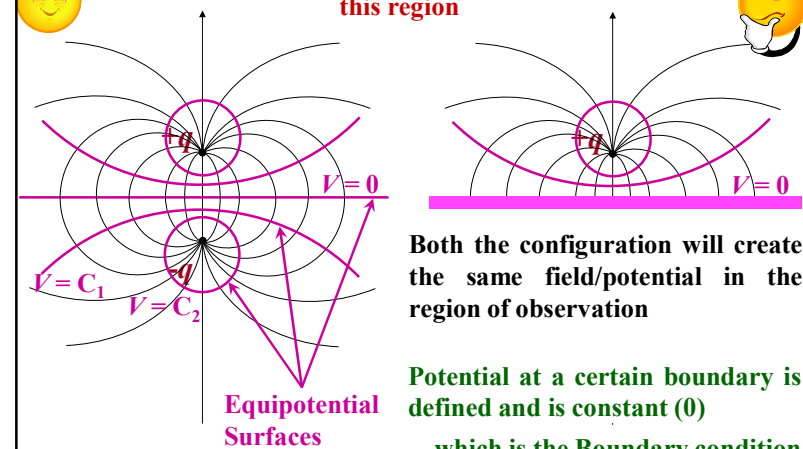
#### What we do...

- 🍁 Solve LE if  $\rho = 0$ , or PE if  $\rho \neq 0$ 
  - Using direct integration if potential is a function of 1 variable
  - Using Separation of Variable if potential is a function of  $> 1$  variable
- 🍁 Apply BC to determine unique solution
- 🍁 Having  $V$ , find the field using  $\vec{E} = -\vec{\nabla}V$
- 🍁 If desired, calculate induced charge density on the conductor

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## Special Techniques in Electrostatics

Nothing (E/V/charge distribution) will change in this region



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**Special Techniques in Electrostatics**

Nothing (E/V/charge distribution) will change in this region

$+q$   $-q$   $V=0$

These two configurations are equivalent in the region of observation

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**Special Techniques in Electrostatics**

$V=0$   $V=C_1$   $V=C_2$

Equipotential Surfaces

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**Special Techniques in Electrostatics**

Nothing (E/V/charge distribution) will change anywhere except to the region within the conductor

$V=0$   $V=C_1$   $V=C_2$

Equipotential Surfaces

Both the configuration will create the same field/potential in the region of observation

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**Special Techniques in Electrostatics**

Nothing (E/V/charge distribution) will change anywhere except to the region within the conductor

$V=0$   $V=C_1$   $V=C_2$

Equipotential Surfaces

Common point in both situation: potential at the boundary of sphere has to be strictly constant

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### Special Techniques in Electrostatics

Nothing (E/V/charge distribution) will change anywhere except to the region within the conductor

Equipotential Surfaces

Potential at a certain boundary is defined and is constant ( $C_2$ )  
...which is the Boundary condition

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### Special Techniques in Electrostatics

Nothing (E/V/charge distribution) will change anywhere except to the region within the conductor

These two configurations are equivalent in the region of observation

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### Special Techniques in Electrostatics

What we have done is quite interesting!

Find the EF/Potential from these configurations at a given point in the region above/outside the conductor

How to solve this problem?

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### Special Techniques in Electrostatics

What we have done is quite interesting!

Find the EF/Potential from these configurations at a given point in the region above/outside the conductor

How to solve this problem?

Is it  $\frac{1}{4\pi\epsilon_0} \frac{q}{r}$ ? NO

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### Special Techniques in Electrostatics

What we have done is quite interesting!

Find the EF/Potential from these configurations at a given point in the region above/outside the conductor

**How to solve this problem?**

Situation is more complicated here!

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### Special Techniques in Electrostatics

What we have done is quite interesting!

Find the EF/Potential from these configurations at a given point in the region above/outside the conductor

**How to solve this problem?**

In both the cases, potential is defined at the conductor, which constitutes BC.

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### Special Techniques in Electrostatics

What we have done is quite interesting!

Find the EF/Potential from these configurations at a given point in the region above/outside the conductor

**How to solve this problem?**

Replace the given system with two equivalent point charges – one original, and another introduced at a proper location within conductor/conductor (image) plane

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### Special Techniques in Electrostatics

What we have done is quite interesting!

Find the EF/Potential from these configurations at a given point in the region above/outside the conductor

**How to solve this problem?**

only condition – point charge system should not alter the boundary potential and actual charge distribution in the region where the field/potential is desired

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## Special Techniques in Electrostatics

What we have done is quite interesting!

### Summary

- Remove the boundary (conductor defines the boundary), which is equipotential.
- Introduce small number of point charge(s) (image charge) of **proper magnitude** at **appropriate location outside the region of interest**.
- This must be done in a way that the individual scalar potential of all charges (**original** as well as **image** charges) must add up to **form an equipotential surface of the same shape and same value** as the **original boundary surface**.



### METHOD OF IMAGES

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## Special Techniques in Electrostatics

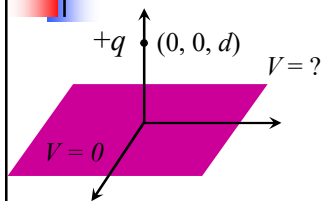
### METHOD OF IMAGES

#### Always remember

- Image charges must be located in the conducting region (outside the region of interest). **IS A MUST** – as this condition is necessary to satisfy Poisson Equation.
- Image charges must be located such that on the conducting surface, potential is zero/constant. **IS A MUST** – as this condition ensures that BCs are satisfied.

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### METHOD OF IMAGES: Example 1



Infinite grounded conducting plane

What we should actually do is like the following:

As charge is given & potential at a given surface is defined (zero here)

We need to solve PE in a region  $z > 0$  with a single point charge at  $(0, 0, d)$  & apply the BC as:

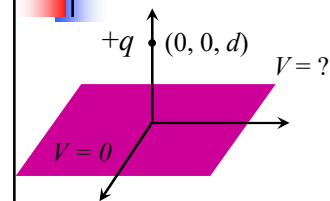
- $V = 0$  at  $Z = 0$  plane
- $V \rightarrow 0$  far away from the charge ( $x^2 + y^2 + z^2 \gg d^2$ )

Uniqueness Theorem guarantees that there will be only one function that meets this requirement

Even by making a guess, if we can discover that function, its got to be the right answer

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### METHOD OF IMAGES: Example 1

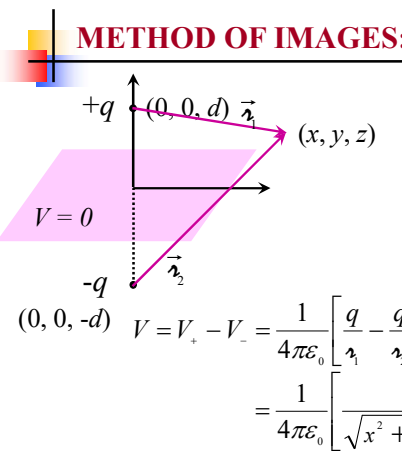


Infinite grounded conducting plane

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**METHOD OF IMAGES: Example 1**



**Potential Calculation**

$$\vec{r}_1 = \sqrt{x^2 + y^2 + (z-d)^2}$$

$$\vec{r}_2 = \sqrt{x^2 + y^2 + (z+d)^2}$$

$$V = V_+ - V_- = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{r_1} - \frac{q}{r_2} \right]$$

$$= \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{\sqrt{x^2 + y^2 + (z-d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z+d)^2}} \right]$$

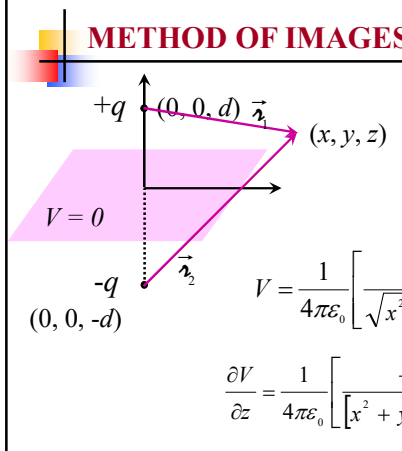
As can be observed from the above function:

- (i)  $V = 0$  when  $z = 0$
- (ii)  $V \rightarrow 0$  when  $(x^2 + y^2 + z^2 \gg d^2)$
- (iii) only charge in the region  $z > 0$  is  $q$  at  $(0, 0, d)$

**BCs are satisfied & this is the only solution**

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**METHOD OF IMAGES: Example 1**



**Induced Charge Density**

$$\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$$

$$\sigma = -\epsilon_0 \frac{\partial V}{\partial z} \Big|_{z=0}$$

$$V = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{\sqrt{x^2 + y^2 + (z-d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z+d)^2}} \right]$$

$$\frac{\partial V}{\partial z} = \frac{1}{4\pi\epsilon_0} \left[ \frac{-q(-d)}{[x^2 + y^2 + (z-d)^2]^{3/2}} + \frac{q(+d)}{[x^2 + y^2 + (z+d)^2]^{3/2}} \right]$$

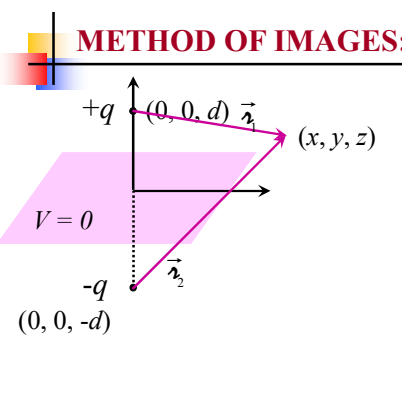
$$\therefore \sigma = \frac{-qd}{2\pi[x^2 + y^2 + d^2]^{3/2}}$$

**Home-Work:** Plot it for  $x = 0$ , and then for  $y = 0$

**Note:** Its not a uniform charge density. Maximum at  $x = y = z = 0$ !

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**METHOD OF IMAGES: Example 1**



**Total Induced Charge**

$$\sigma = \frac{-qd}{2\pi[x^2 + y^2 + d^2]^{3/2}}$$

$$Q = \int \sigma da$$

$$= \iint \frac{-qd dx dy}{2\pi(x^2 + y^2 + d^2)^{3/2}}$$

$$= \int_{r=0}^{\infty} \int_{\phi=0}^{2\pi} \frac{-qd r dr d\phi}{2\pi(r^2 + d^2)^{3/2}}$$

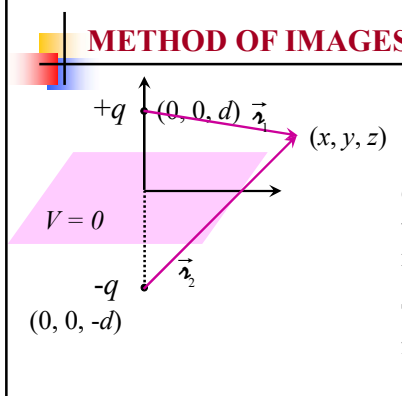
$$= \frac{-qd}{2} \int_{r=0}^{\infty} \frac{2r dr}{(r^2 + d^2)^{3/2}}$$

$$= qd \left[ \frac{1}{\infty} - \frac{1}{d} \right] = -q$$

**As expected!**

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**METHOD OF IMAGES: Example 1**



**Force**

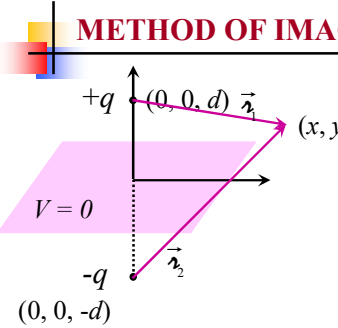
Charge  $q$  would be attracted towards the plane because of the induced -ve charge.

This force would be same as the force experience by  $q$  due to  $-q$ .

$$\vec{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{z}$$

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### METHOD OF IMAGES: Example 1



**Energy**

With two point charges ( $q, -q$ ) and no conductor, it would be

$$W = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{2d}$$

Let us calculate energy more rigorously.

$$W = \int_{-\infty}^d \vec{F} \cdot d\vec{l} = \frac{1}{4\pi\epsilon_0} \int_{-\infty}^d \frac{q^2}{4z^2} \hat{z} \cdot dz \hat{z} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d} \quad \vec{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{z}$$

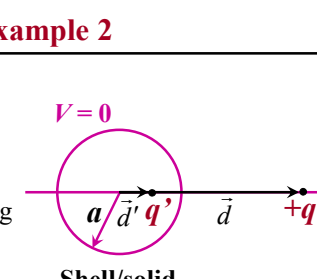
Half of what METHOD OF IMAGES suggests!

What's wrong?

Nothing

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### METHOD OF IMAGES: Example 2



BC: Potential at the conducting surface is zero.

**Image charge:**

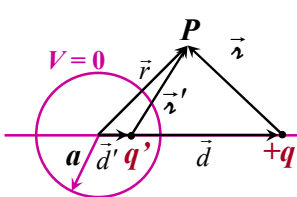
Must be in conductor's plane, i.e., inside the conductor.

Magnitude and location ( $q', d'$ ) of the image charge must be chosen such that the spherical surface at  $r = a$  (in the absence of the conductor) becomes an equipotential with  $V = 0$ .

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### METHOD OF IMAGES: Example 2

**Potential at P:**

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|\vec{r} - \vec{d}|} + \frac{q'}{|\vec{r} - \vec{d}'|} \right] \quad r > a$$


**Applying BC at  $r = a$**

$$V(\vec{r})|_{r=a} = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|a\hat{r} - d\hat{d}|} + \frac{q'}{|a\hat{r} - d'\hat{d}|} \right] = 0$$

$$= \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|a\hat{r} - d\hat{d}|} + \frac{q'}{\frac{a}{d} |d\hat{r} - dd'\hat{d}|} \right] = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|a\hat{r} - d\hat{d}|} + \frac{(d/a)q'}{|d\hat{r} - \frac{dd'}{a}\hat{d}|} \right] = 0$$

If  $d' = a^2 / d$

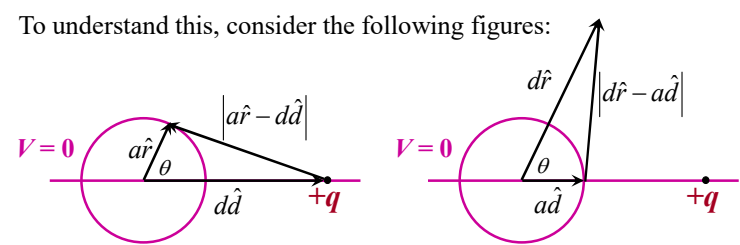
$$V(\vec{r})|_{r=a} = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|a\hat{r} - d\hat{d}|} + \frac{(d/a)q'}{|d\hat{r} - a\hat{d}|} \right] \quad \text{Here, always } |a\hat{r} - d\hat{d}| = |d\hat{r} - a\hat{d}|$$

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### METHOD OF IMAGES: Example 2

$$V(\vec{r})|_{r=a} = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|a\hat{r} - d\hat{d}|} + \frac{(d/a)q'}{|d\hat{r} - a\hat{d}|} \right] \quad \text{Here, always } |a\hat{r} - d\hat{d}| = |d\hat{r} - a\hat{d}|$$

To understand this, consider the following figures:



Using cosine law:  $|a\hat{r} - d\hat{d}| = \sqrt{a^2 + d^2 - 2ad \cos \theta}$

$|d\hat{r} - a\hat{d}| = \sqrt{d^2 + a^2 - 2da \cos \theta}$

$|a\hat{r} - d\hat{d}|$  will always be equal to  $|d\hat{r} - a\hat{d}|$

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### METHOD OF IMAGES: Example 2

Potential at  $P$ :

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|\vec{r} - \vec{d}|} + \frac{q'}{|\vec{r} - \vec{d}'|} \right] \quad r > a$$

Applying BC at  $r = a$

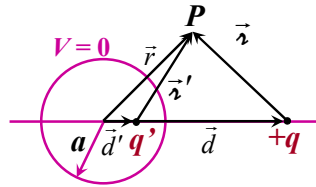
$$V(\vec{r})|_{r=a} = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|a\hat{r} - d\hat{d}|} + \frac{q'}{|a\hat{r} - d'\hat{d}|} \right] = 0$$

$$= \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|a\hat{r} - d\hat{d}|} + \frac{q'}{\frac{a}{d} |d\hat{r} - dd'\hat{d}|} \right] = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|a\hat{r} - d\hat{d}|} + \frac{(d/a)q'}{|d\hat{r} - dd'\hat{d}|} \right] = 0$$

If  $d' = a^2/d$

$$V(\vec{r})|_{r=a} = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|a\hat{r} - d\hat{d}|} + \frac{(d/a)q'}{|d\hat{r} - ad\hat{d}|} \right] \quad \text{This will be zero iff } q = -\left(\frac{d}{a}\right)q'$$

$$q' = -\left(\frac{a}{d}\right)q \quad d' = \frac{a^2}{d}$$



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### METHOD OF IMAGES: Example 2

Potential at  $P$ :

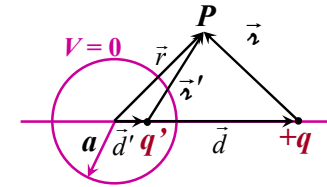
$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|\vec{r} - \vec{d}|} + \frac{q'}{|\vec{r} - \vec{d}'|} \right] \quad r > a$$

$$q' = -\left(\frac{a}{d}\right)q$$

$$d' = \frac{a^2}{d}$$

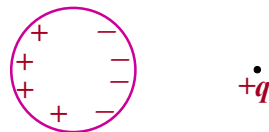
We can now calculate  $E$ ,  $F$ ,  $q$ , and  $\sigma$

$$\vec{F} \text{ (on } q \text{ by an image charge)} = \frac{qq'}{4\pi\epsilon_0(d-d')^2} = \frac{-aq^2\vec{d}}{4\pi\epsilon_0(d^2 - a^2)^2}$$



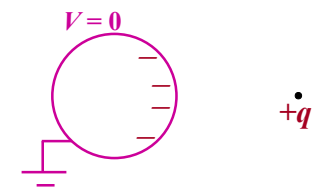
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### METHOD OF IMAGES: Example 2



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### METHOD OF IMAGES: Example 2



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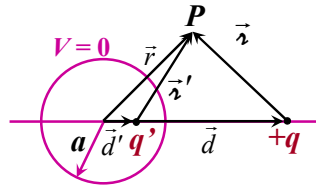
### METHOD OF IMAGES

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|\vec{r} - \vec{d}|} + \frac{q'}{|\vec{r} - \vec{d}'|} \right] \quad r > a$$

$$q' = -\left(\frac{a}{d}\right)q$$

$$d' = \frac{a^2}{d}$$

The solution for the point charge outside a grounded sphere can be used as a starting point to find the solution of several related problems



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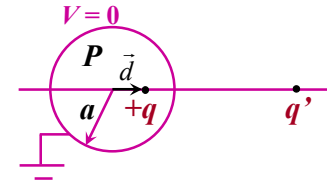
### METHOD OF IMAGES: Example 3

A point charge is kept inside a grounded hollow conducting sphere. Find potential within the sphere?

Solution will be same as before as the solution of the previous problem never demanded  $d > a$ !

Image charge will now be outside the sphere

Solution will be valid now within the space of the hollow sphere



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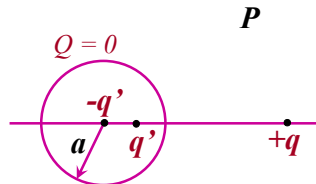
### METHOD OF IMAGES: Example 4

A point charge outside an isolated, neutral conducting sphere.

BC: Net charge on the conducting surface is defined.

Starting point: Example 2.

- Start with a grounded conducting sphere in front of  $q$ .  $q'$  charge would be distributed over its surface.
- Disconnect the grounded wire.
- Add to the sphere a charge  $-q'$  to make total charge zero and the conductor neutral.
- This added charge  $-q'$  will distribute itself uniformly over the surface as the electrostatic force due to the point charge  $q$  is already balanced by the charge  $q'$ .
- Potential due to the added charge  $-q'$  will be the same as if a point charge of this magnitude is kept at the origin



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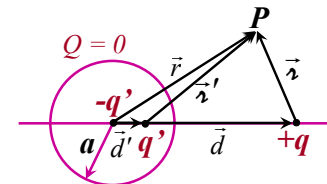
### METHOD OF IMAGES: Example 4

A point charge outside an isolated, neutral conducting sphere.

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{|\vec{r} - \vec{d}|} + \frac{q'}{|\vec{r} - \vec{d}'|} + \frac{-q'}{|\vec{r}|} \right] \quad r > a$$

$$q' = -\left(\frac{a}{d}\right)q$$

$$d' = \frac{a^2}{d}$$



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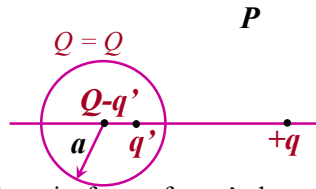
### METHOD OF IMAGES: Example 5

A point charge outside an isolated, charged conducting sphere.

BC: Net charge on the conducting surface is defined.

Starting point: Example 2.

- Start with a grounded conducting sphere in front of  $q$ .  $q'$  charge would be distributed over its surface.
- Disconnect the grounded wire.
- Add to the sphere a charge  $Q - q'$  to make total charge zero and the conductor neutral.
- This added charge  $Q - q'$  will distribute itself uniformly over the surface as the electrostatic force due to the point charge  $q$  is already balanced by the charge  $q'$ .
- Potential due to the added charge  $Q - q'$  will be the same as if a point charge of this magnitude is kept at the origin.

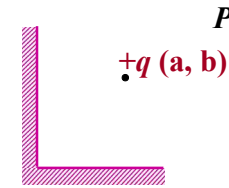


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### METHOD OF IMAGES: Example 6

Till now we have encountered only one conductor surface as the boundary for the problem.

Consider a point charge in front of two semi-infinite grounded conductor planes at right angle. Find  $V$  in the region between the two conductors where charge is located.

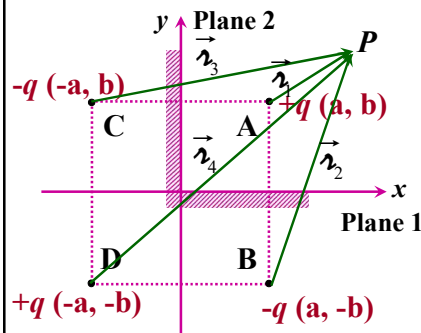


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### METHOD OF IMAGES: Example 6

Till now we have encountered only one conductor surface as the boundary for the problem.

Consider a point charge in front of two semi-infinite grounded conductor planes at right angle. Find  $V$  in the region between the two conductors where charge is located.



$$V = \frac{q}{4\pi\epsilon_0} \left[ \frac{1}{r_1} - \frac{1}{r_2} - \frac{1}{r_3} + \frac{1}{r_4} \right]$$

$$r_1 = [(x-a)^2 + (y-b)^2 + z^2]^{1/2}$$

$$r_2 = [(x-a)^2 + (y+b)^2 + z^2]^{1/2}$$

$$r_3 = [(x+a)^2 + (y-b)^2 + z^2]^{1/2}$$

$$r_4 = [(x+a)^2 + (y+b)^2 + z^2]^{1/2}$$

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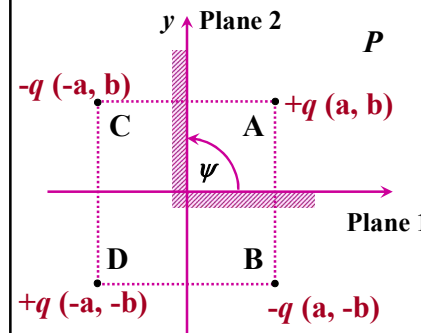
### METHOD OF IMAGES: Example 6

Consider a point charge in front of two semi-infinite inclined grounded conductor planes. Find  $V$  in the region between the two conductors where charge is located.

#### Important Note

If the angle of inclination  $\psi$  is not an integer divisor of  $180^\circ$ , image charges can not satisfy the BC ( $V=0$  at conductor)

In this case image charge will not remain only in the conducting/image plane, but will be in the region of interest.



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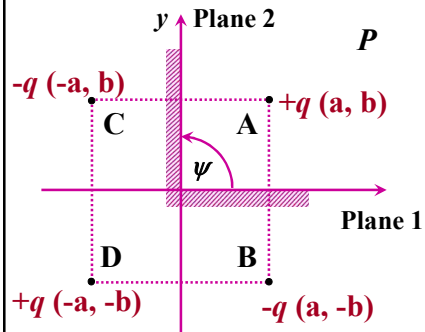
### METHOD OF IMAGES: Example 6

Consider a point charge in front of two semi-infinite inclined grounded conductor planes. Find  $V$  in the region between the two conductors where charge is located.

#### Important Note

For the angle of inclination  $\psi$  being an integer divisor of  $180^\circ$ , total number of images would be

$$N = \left( \frac{360^\circ}{\psi} - 1 \right)$$



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### METHOD OF IMAGES: Example 6

Consider a point charge in front of two semi-infinite inclined grounded conductor planes. Find  $V$  in the region between the two conductors where charge is located.

#### Important Note

Consider CPCS with z-axis along the line of intersection of the conductor surfaces.

Coordinates of  $q$  would be  $(\rho, \phi)$ .

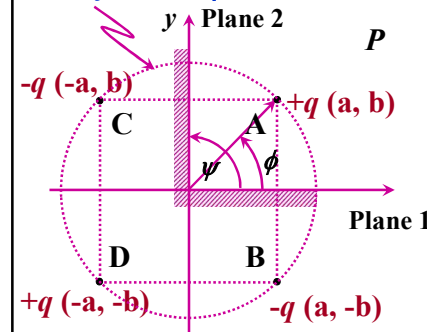
**Reflection through Plane 1:**

$$q(\rho, \phi) \rightarrow -q(\rho, -\phi)$$

**Reflection through Plane 2:**

$$q(\rho, \phi) \rightarrow -q(\rho, 2\psi - \phi)$$

Image charges on the circle of constant  $\rho$



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